VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME			
CLASS		TUTOR NAME	

PHYSICS 9478/03

Paper 3 Longer Structured Questions **18 September 2026**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and tutor name in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

Section A
Answer **all** questions.

Section B
Answer **one** question only.

You are advised to spend one and a half hours on Section A and half an hour on Section B.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 8
2	/ 7
3	/ 9
4	/ 5
5	/ 9
6	/ 9
7	/ 8
8	/ 20
9	/ 20
Total	/ 75

This document consists of **27** printed pages.

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} $(1 / (36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = F / A$
gravitational potential	$\phi = -Gm / r$
temperature	$T / \text{K} = T / ^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal molecule	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$
charging a capacitor	$Q = Q_0 \left[1 - e^{-\frac{t}{\tau}} \right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2} n^2$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in the spaces provided.

1 A ball is thrown horizontally from the top of a building, as shown in Fig. 1.1.

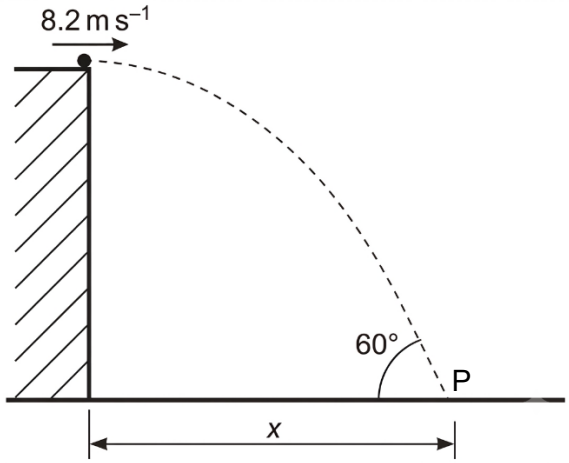


Fig. 1.1

The ball is thrown with a horizontal speed of 8.2 m s^{-1}. The side of the building is vertical. The ball hits the ground at point P at a distance *x* from the building and at an angle of 60° to the horizontal. Air resistance is negligible.

(a) For the ball at point P,

(i) show that the vertical component of its velocity is 14 m s^{-1},

$$\tan\theta = \frac{v_y}{v_x} \qquad [1]$$

$$v_y = v_x \tan\theta$$
$$= 8.2\tan 60°$$
$$= 14.2 \quad [1]$$
$$= 14 \text{ m s}^{-1}$$

[2]

(ii) determine the vertical distance it has fallen,

$$v_y^2 = u_y^2 + 2a_y s_y$$
$$14^2 = 0 + 2(9.81)s_y \quad \text{(Take downwards as +ve)} \qquad [1]$$
$$s_y = 9.99$$
$$= 10 \text{ m (2 sf)} \qquad [1]$$

vertical distance = ... m [2]

(iii) determine the horizontal distance *x*.

Let t be the time taken to fall through the height:

$v_y = u_y + a_y t$

$14 = 0 + 9.81t$

$t = 1.43$ s [1]

Then $s_x = u_x t$

$= 8.2 \times 1.43$

$= 11.7$

$= 12$ m (2 sf) [1]

distance *x* = ... m [2]

(b) State and explain the effect on your answer to the horizontal distance *x* if air resistance is not negligible.

Air drag causes the horizontal velocity to continually decrease throughout the motion. [1]

Because the average horizontal speed during the ball's flight is lower, the ball covers less horizontal distance in a given time [1] [2]

[Total: 8]

2 **(a)** State Newton's second law of motion.

The rate of change of momentum of an object is directly proportional to the resultant force acting on it, and takes place in the direction of the force. [1]

(b) A ball of mass 150 g travelling horizontally at a speed of 20 m s^{-1} collides with a vertical wall. Fig. 2.1 shows how the force of impact on the wall, *F*, varies with time *t*.

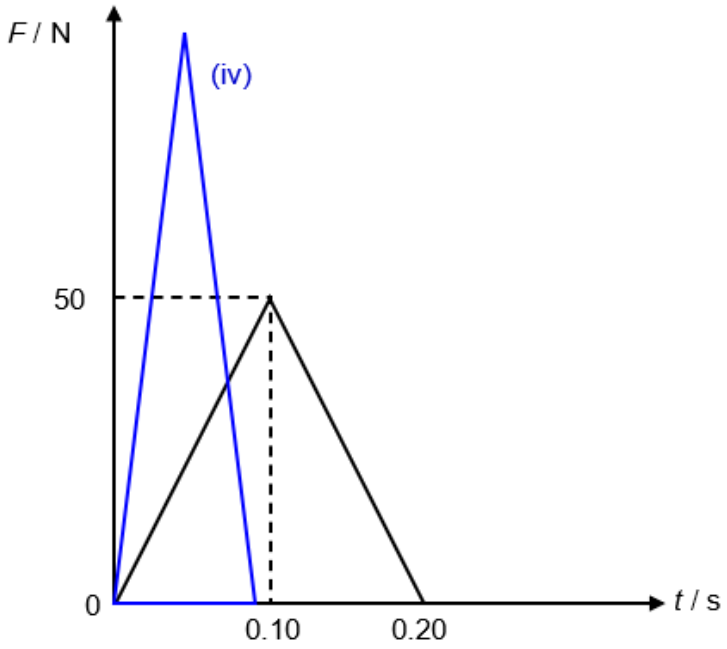


Fig. 2.1

(i) Determine the change in momentum of the ball.

Change in momentum = area under graph [1]

$$= \frac{50 \times 0.20}{2}$$

= 5.0 N s [1]

change in momentum = ... N s [2]

(ii) Hence determine the final speed of the ball.

Take rightwards as positive.
(Note: the impact force on the ball is to the left. So change in momentum of the ball is also to the left i.e. negative.)

Change in momentum = $m(v_f - v_i)$

$$-5.0 = 0.150(v_f - 20)$$

$$v_f = -13.33 \text{ m s}^{-1}$$

$\therefore$ Speed = 13 m s^{-1}

final speed = ... m s^{-1} [1]

(iii) Balls which are identical to the one above are thrown at the wall at the rate of 30 balls per minute. Determine the average impact force on the wall.

$$\text{Average force} = \frac{\text{Total change in momentum of 30 balls}}{\text{time interval}} \quad [1]$$

$$= \frac{30 \times 5.0}{60}$$

$$= 2.5 \text{ N} \quad [2]$$

average impact force = ... N [2]

(iv) A ball which is equal in mass as the one above is made of a harder material. It also collides with the wall at the same speed as before. Draw on Fig. 2.1 a second graph to show how the impact force on the wall varies with time when this ball collides with the wall. [1]

Higher peak and shorter time interval [1]

[Total: 7]

3 A sound system in a large auditorium has two identical loudspeakers, S_1 and S_2, separated by 1.5 m. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage across its terminals. A balance control is used to adjust the voltage supplied to each loudspeaker. The arrangement is shown in Fig. 3.1.

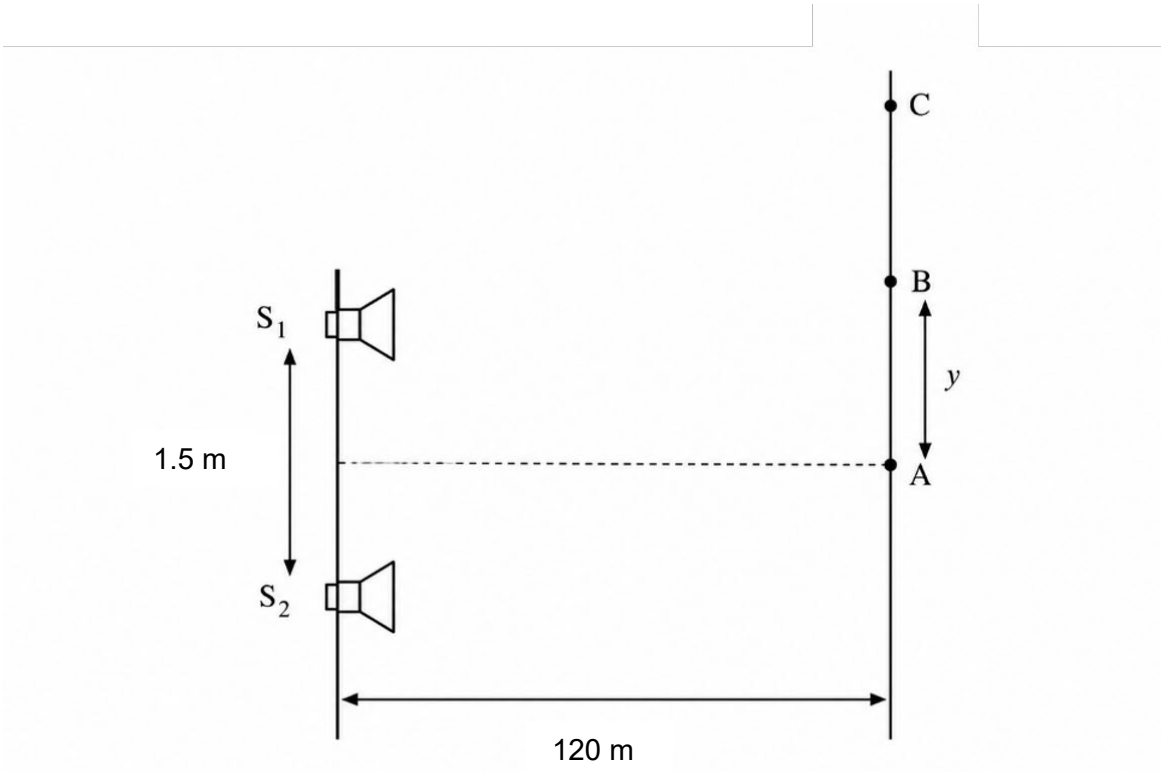


Fig. 3.1 (not to scale)

Initially, the loudspeakers emit sound waves of frequency 8500 Hz in phase. The balance control is adjusted so that the potential difference across each loudspeaker is 5.0 V r.m.s.

An observer at A hears a loud sound of intensity I_{max}. As he moves along the line AC, which is 120 m from the loudspeakers, the intensity first falls to zero at B, a distance y from A.

The speed of sound in air is 340 m s^{-1}.

(a) Determine the distance y.

$$\lambda = \frac{v}{f} = \frac{340}{8500} = 0.0400\,\text{m} \qquad [1]$$

Let the distance between the first two maxima be x.

$$x = \frac{\lambda D}{D} = \frac{0.0400 \times 120}{1.5} = 3.2\,\text{m} \quad [1]$$

Since B is the first minimum, the distance y is half of x.

$$y = \frac{x}{2} = \frac{3.2}{2} = 1.6\,\text{m} \quad [1]$$

y = ... m [3]

(b) Determine the next higher frequency of sound emitted by the speaker such that point B would also be a point of zero intensity.

The path difference at B is $\Delta x = \frac{\lambda}{2} = \frac{0.0400}{2} = 0.0200\,\text{m}$ [1]

For destructive interference, $\Delta x = (n + \frac{\lambda}{2})$

For the original frequency, this corresponds to n = 0. If the frequency is increased, it will correspond to n = 1 or $\Delta x = \frac{3\lambda'}{2}$

Hence

$0.0200 = \frac{3\lambda'}{2}$

$\lambda' = 0.01333\,\text{m}$ [1]

$f' = \frac{v}{\lambda'} = \frac{340}{0.01333} = 2.55 \times 10^4\,\text{Hz}$ [1]

frequency = ... Hz [3]

(c) The loudspeakers continue to emit sound of frequency 8500 Hz. The balance control is adjusted so that the voltages applied across S_1 and S_2 are 2.0 V r.m.s. and 8.0 V r.m.s. respectively. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage applied across it.

In terms of I_{max}, determine the new intensity at B.

Initially, resultant amplitude at A, $A = A_1 + A_2 = 5 + 5 = 10$ units [1]

Later, the balance control is changed. At B, the waves meet out of phase:
$\therefore A' = A_1 - A_2 = 8 - 2 = 6$ units [1]

$\frac{I_{max}'}{I_{max}} = \frac{kA'^2}{kA^2} = (\frac{6}{10})^2$

$\therefore I_{max}' = 0.36 I_{max}$ [1]

Intensity = ...I_{max} [3]

Total: [9]

4 **(a)** Define *electric field strength*.

Electric field strength at a point is defined as the **electric force per unit positive charge** exerted on a **small test charge** placed at that point.

.....[1]

(b) An oil drop of mass 1.0×10^{-15} kg and carrying a charge of 1.6×10^{-19} C is placed in the middle of two horizontal parallel plates.

The separation between the plates is 40 mm and the the potential difference between the plates is 8.8 kV. The oil drop is seen to be attracted towards the upper plate because of its electric charge.

(i) Calculate the electric force experience by the oil drop.

Electric force = qE

$$= q\frac{V}{d} \quad [1]$$

$$= \left(1.6\times10^{-19}\right)\frac{8.8\times10^{3}}{40\times10^{-3}}$$

$$= 3.52\times10^{-14} \text{ N} \quad [1]$$

force = ... N [2]

(ii) Initially the oil drop is 20 mm from either plate and is at rest. Ignoring air resistance, use energy considerations to calculate its speed when it reaches the upper plate.

By Conservation of Energy,
Gain in kinetic energy + Gain in GPE = Loss in electrical potential energy

$$\frac{1}{2}mv^2 - 0 = q\Delta V + mg\Delta h \quad [1]$$

$$\frac{1}{2}\left(1.0\times10^{-15}\right)v^2 = \left(1.6\times10^{-19}\right)\left(\frac{8.8\times10^{3}}{2}\right) + \left(1.0\times10^{-15}\right)\times9.81\times(20\times10^{-3})$$

$$v = 1.01 \text{ m s}^{-1} \quad [1]$$

speed = ... m s^{-1} [2]

[Total: 5]

5 During cosmic-ray experiments in the 1930s, researchers observed the track of a previously unknown positively charged particle in a cloud chamber. The particle was measured to have a charge of +1 *e* and is known as the muon.

The particle enters a uniform magnetic field of flux density 0.250 T with a speed of 2.40×10^{7} m s^{-1}. Its velocity makes an angle of 60° with the direction of the magnetic field, causing it to follow a helical path as seen in Fig. 5.1 below.

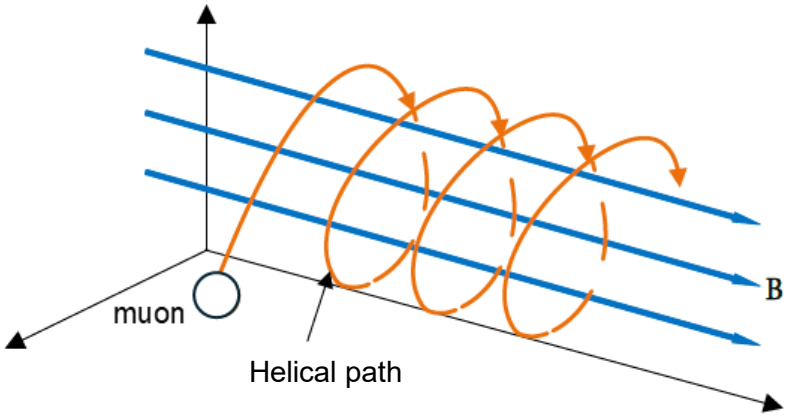


Fig. 5.1

Measurements of the track show that the muon took 2.96×10^{-8} s to complete one revolution.

(a) (i) Show that the radius of the helical path is 0.0979 m.

$$v_{\perp} = r\omega = r\frac{2\pi}{T}$$

$$r = \frac{v_{\perp}T}{2\pi}$$

$$= \frac{2.40\times10^{7}\sin60^{\circ}\times2.96\times10^{-8}}{2\pi}$$

$$= 0.09792$$

$$\approx 0.979 \text{ m}$$

[2]

(ii) Determine the mass, m_{μ} of the muon.

The magnetic force provides the centripetal force:

$$Bqv_{\perp} = \frac{mv_{\perp}^{2}}{r}$$ [1]

$$m = \frac{Bqr}{v_{\perp}}$$

$$= \frac{0.250\times1.60\times10^{-19}\times0.0979}{2.40\times10^{7}\sin60^{\circ}}$$

$$= 1.88\times10^{-28} \text{ m}$$ [1]

m_{μ} = ... kg [2]

(b) Determine the distance travelled by the particle **parallel** to the magnetic field during one complete revolution.

Distance moved in 1 revolution

$d = v_{\parallel} T$

$= v \cos 60^{\circ} T$ [1 for correct resolution]

$= 2.4 \times 10^{7} \times \cos 60^{\circ} \times 2.96 \times 10^{-8}$

$= 0.355$ m [1]

distance = ... m [2]

(c) Before taking measurements, the researchers compensate for the vertical component of the Earth's magnetic field.

A short magnetic needle is suspended from a string 5.0 cm from a horizontal long straight wire, as shown in the top view in Fig. 5.2.

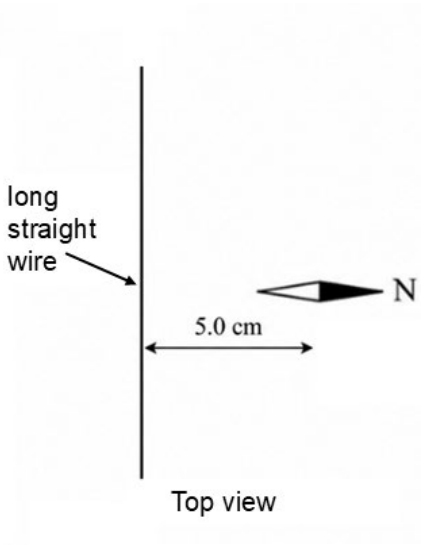


Fig. 5.2

When there is no current in the wire, the north pole of the needle dips at an angle of 6.0° below the horizontal, as shown in the side view in Fig. 5.3.

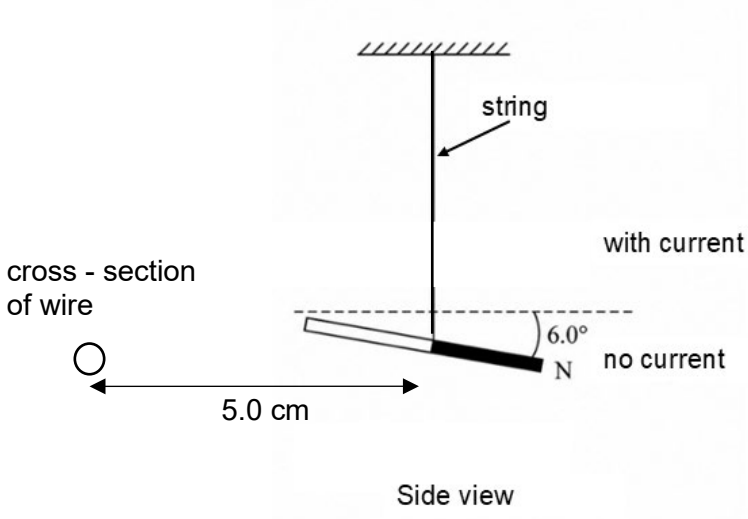


Fig. 5.3

A current of 0.48 A is then passed through the wire. The magnetic field produced by the current causes the needle to become horizontal, as shown in the side view in Fig. 5.4.

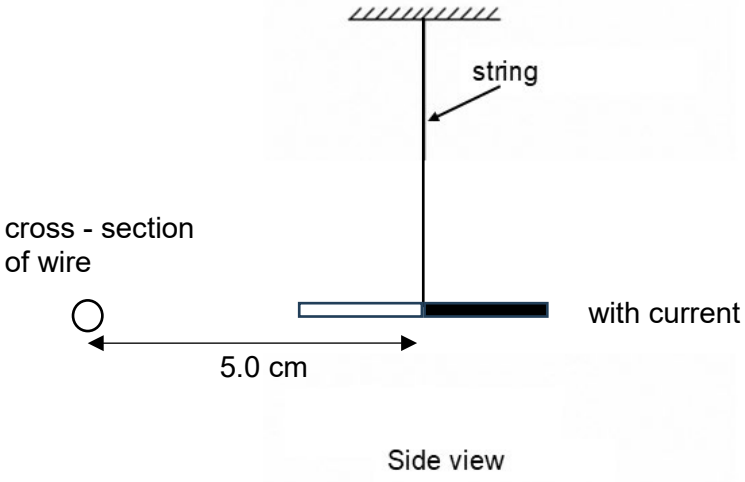


Fig. 5.4

(i) Show that the magnetic flux density produced by the wire, when the current is flowing through it, at the position of the short magnetic needle is 1.9×10^{-6} T.

The magnetic field due to a long straight wire is

$$B_{\text{wire}} = \frac{\mu_0 I}{2\pi r}$$

$$= \frac{(4\pi \times 10^{-7})(0.47)}{2\pi(0.050)}$$

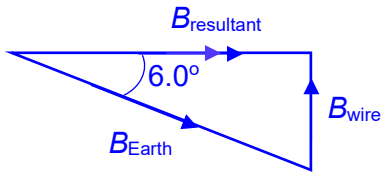
$$= 1.88 \times 10^{-6} \text{ T} \quad [1]$$

$$\approx 1.9 \times 10^{-6} \text{ T} \quad [1]$$

(ii) Determine the magnitude of the Earth's magnetic flux density in the laboratory.

The Earth's magnetic field makes an angle of 6.0º below the horizontal.
The wire's field is vertical at the position of the needle, so for the needle to become horizontal, the vertical component of the Earth's field must be cancelled.

$$B_{wire} = B_{Earth} \sin 6.0^\circ \qquad [1]$$



$$B_{Earth} = \frac{1.88 \times 10^{-6}}{\sin 6.0^\circ}$$
$$= 1.80 \times 10^{-5} \text{ T} \qquad [1]$$

magnetic flux density = ...T [2]

Total: [9]

6 A railway depot uses an electrical de-icing unit to prevent ice from forming on a track-switch mechanism. The unit contains a resistive heating element supplied by an ideal step-down transformer. A diagram of the setup is shown in Fig. 6.1 below.

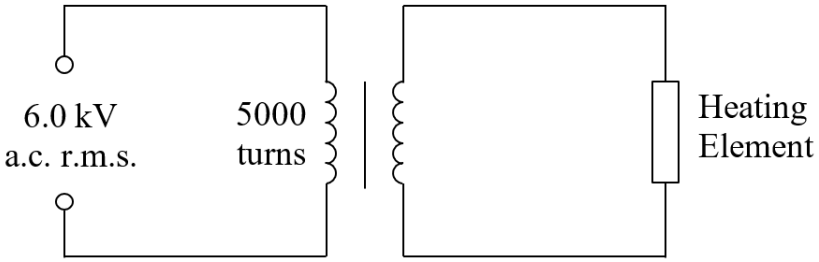


Fig. 6.1

The primary coil of the transformer is connected to a 6.0 kV, 50 Hz sinusoidal alternating supply. The potential difference across the secondary coil is 240 V r.m.s.. The primary coil has 5000 turns.

(a) Determine the number of turns on the secondary coil.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$N_s = 5000\left(\frac{240}{6000}\right)$$

$$= 200 \text{ turns} \qquad [1]$$

number of turns= ..... [1]

(b) The heating element behaves as a pure resistor and dissipates a mean power of 4.6 kW. The instantaneous current I in the heating element is represented by

$$I = I_o \sin \omega t \, .$$

Determine I_o and ω.

$$P_{\text{mean}} = V_{rms} I_{rms}$$

$$I_{rms} = \frac{4600}{240} = 19.17 \text{ A} \qquad [1]$$

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

$$I_0 = \sqrt{2} \times 19.17 = 27.1 \text{ A} \qquad [1]$$

$$\omega = 2\pi f = 2\pi(50) = 314 \text{ rad s}^{-1} \qquad [1]$$

I_o = ... A

ω = ... rad s^{-1} [3]

(c) Determine the r.m.s. current in the primary coil of the transformer.

$V_p I_p = V_s I_s$

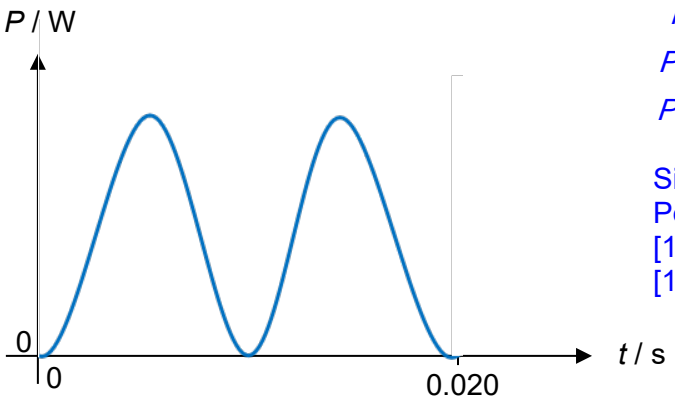
$I_p = \dfrac{V_s I_s}{V_p}$

$= \dfrac{(240)(19.17)}{6000}$ [1]

$= 0.767$ A [1]

current = ... A [2]

(d) On Fig. 6.2, sketch the variation with time t of the power P dissipated in the resistor from $t = 0$ to $t = 0.020$ s.



$I = I_0 \sin \omega t$

$P = I^2 R$

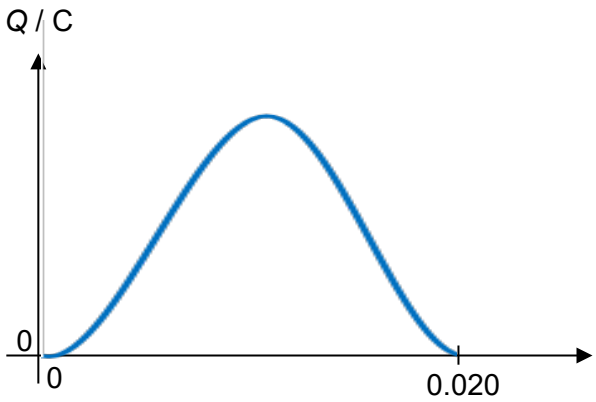
$P \propto \sin^2 \omega t$

Since the frequency is 50 Hz, the Period of the sine function is 0.02 s.
[1] for correct shape
[1] for correct period

Fig. 6.2

[2]

(e) On Fig. 6.3, sketch the variation of the net amount of charge Q which flowed past a point in the resistor from $t = 0$ to $t = 0.020$ s. (Numerical values for the charge are not expected).



The gradient of the Q-t graph gives the I-t graph, which is a sine graph. The Q-t graph has a maximum occurring at half of the period of the sine function. After half a period, the current reverses, so the charges flow in the opposite direction and the net charge flowed decreases until zero.
[1] correct shape

Fig. 6.3

[1]

[Total: 9]

7 **(a)** Radioactive decay is both *random* and *spontaneous*. State what is meant by:

(i) random

A random process is one in which it cannot be predicted which nucleus will decay at a particular instant or when any particular nucleus will decay. [1]

(ii) spontaneous

A spontaneous process is one which is not triggered or affected by external factors such as temperature and pressure. [1]

(b) Polonium-193 decays into lead-189 and an alpha particle. The binding energy per nucleon of the particles are given in Table 7.1.

particle	binding energy per nucleon / MeV
$^{193}_{84}Po$	7.774
$^{189}_{82}Pb$	7.826
$^{4}_{2}He$	7.074

Table 7.1

(i) Determine the energy, in MeV, released when a nucleus of polonium-193 decays into a nucleus of lead-189.

energy released = B.E. of Pb-189 + B.E. of helium – B.E. of Po-193

= (189 x 7.826) + (4 x 7.074) – (193 x 7.774) [1]

= 7.028 = 7.03 MeV [1]

energy = ... MeV [2]

(ii) At an instance in time, a block of polonium was found to be emitting energy at a rate of 16.1 W. Calculate the activity of the polonium block.

Energy released per decay in J =$7.028 \times 10^{6} \times 1.60 \times 10^{-19} = 1.12448 \times 10^{-12}$ J [1]

Power emitted = no. of decays per second x energy per decay

Activity = no. of decays per second = power / energy per decay

$= 16.1 / (1.12448 \times 10^{-12})$

$= 1.4318 \times 10^{13}$

$= 1.43 \times 10^{13}$ Bq [1]

activity = ... Bq [2]

(iii) The half-life of polonium-193 is approximately 420 ms. Calculate the time it will take a pure sample of polonium to decay to 1.0% of its initial amount.

$$\text{Decay constant } \lambda = \frac{\ln 2}{\tau_{1/2}}$$
$$= \frac{\ln 2}{420 \times 10^{-3}}$$
$$= 1.6504 \text{ s}^{-1} \quad [1]$$

$$N = N_0 e^{-\lambda t}$$
$$\text{Tme taken } t = \frac{\ln\left(\frac{N}{N_0}\right)}{-\lambda}$$
$$= \frac{\ln 0.01}{-1.6504}$$
$$= 2.79 \text{ s} \quad [1]$$

time = ... s [2]

[Total: 8]

Section B

Answer **one** question from this section in the spaces provided.

8 A theme-park ride consists of two cages. They are moving in a circular path at constant speed v about a horizontal axis. Fig. 8.1 shows the ride at one instant when cage A is vertically above cage B.

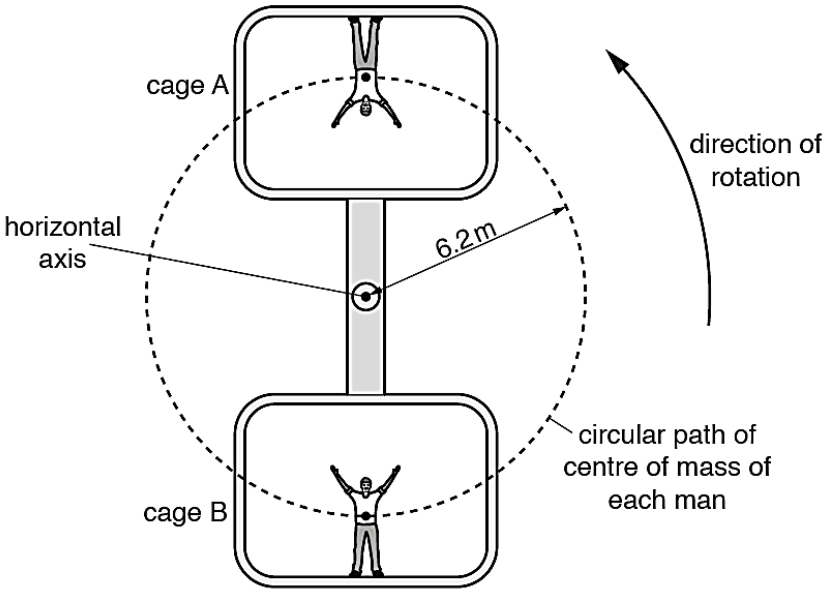


Fig. 8.1

(a) A man is riding in each cage. The mass of each man is 75 kg. The centre of mass of each man is 6.2 m from the horizontal axis. The period of one rotation is 4.1 s.

(i) Determine the speed v of the centre of mass of each man.

$$v = r\omega$$
$$= r\left(\frac{2\pi}{T}\right)$$
$$= 6.2\left(\frac{2\pi}{4.1}\right) \quad [1]$$
$$= 9.5 \text{ m s}^{-1} \quad [1]$$

speed v = ... m s^{-1} [2]

(ii) Calculate the magnitude of the acceleration of the centre of mass of each man.

$$a = r\omega^2$$
$$= r\left(\frac{2\pi}{T}\right)^2$$
$$= 6.2\left(\frac{2\pi}{4.1}\right)^2 \quad [1]$$
$$= 14.6 \text{ m s}^{-2} \quad [1]$$

acceleration = ... m s^{-2} [2]

(b) Fig. 8.2 shows the forces acting on the man in cage B at the instant the cages are in the positions shown. It shows the man in cage A at that same instant.

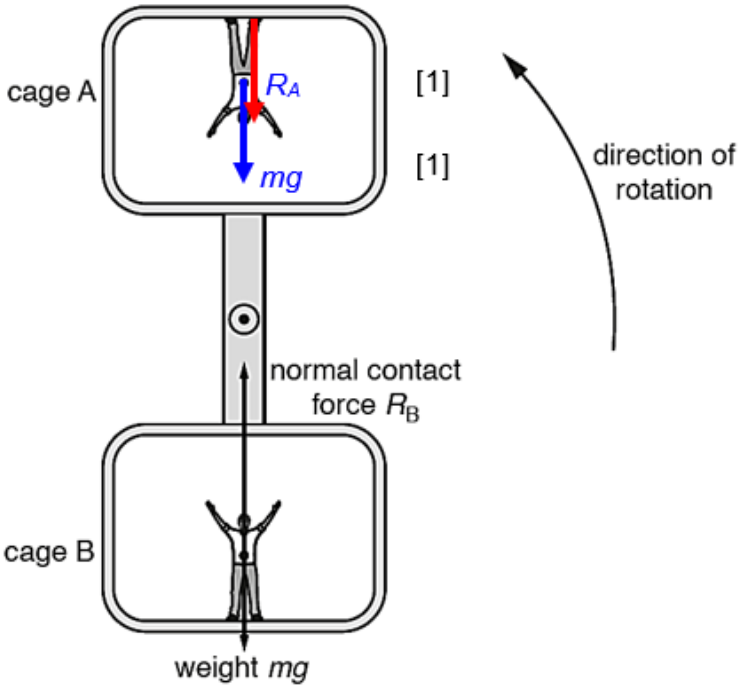


Fig. 8.2

(i) On Fig. 8.2, draw labelled arrows to represent the magnitude and direction of the forces acting on the man in cage A. [2]

(ii) Calculate the magnitude of the normal contact force R_A on the man in cage A at this instant.

$mg + R_A = ma_c$ [1]

$R_A = ma_c - mg$

$= 75(14.6 - 9.81)$

$= 360$ N (2 sf) [1]

R_A = ... N [2]

(c) **(i)** Explain why a minimum value for the speed is needed for the man in cage A to maintain in contact with the floor of his cage.

$R = ma - mg$, $R > 0$ is required to remain contact. [1]

so $a > g$, i.e. $v^2 / r > g$

$v > \sqrt{rg}$, the minimum speed for a contact force to be required [1] [2]

(ii) Calculate this minimum speed.

From (i), $v_{min} = \sqrt{rg}$

$= \sqrt{6.2 \times 9.81}$

$= 7.8$ m s^{-1} [1]

minimum speed = ... m s^{-1} [1]

(d) (i) State Newton's Law of Gravitation.

NLG states that two point masses attract each other with a force directly proportional to the product of the masses [1], and inversely proportional to the square of the distance between them. [1] [2]

(ii) The astronomer Johannes Kepler showed that the period T of rotation of a planet about the Sun is related to its mean distance R from the centre of the Sun by the expression

$$\frac{R^3}{T^2} = k$$

where k is a constant.

Show that, for planets in circular orbits about the Sun of mass M, the constant k is given by

$$k = \frac{GM}{4\pi^2}$$

where G is the gravitation constant.

Gravitational force provides the centripetal force [1]

$$\frac{GMm}{R^2} = mR\omega^2 \quad [1]$$

$$\frac{GMm}{R^2} = mR\left(\frac{2\pi}{T}\right)^2 \quad [1]$$

$$\frac{R^3}{T^2} = \frac{GM}{4\pi^2}$$

[3]

(iii) Earth can be assumed to be in a circular orbit about the Sun with an orbital radius of 1.50×10^8 km.

Use the expression in **d(ii)** to calculate a value for the mass of the Sun.

$$\frac{R^3}{T^2} = \frac{GM}{4\pi^2}$$

$$\frac{(1.50\times10^{11})^3}{(365\times24\times60\times60)^2} = \frac{6.67\times10^{-11}\times M}{4\pi^2} \quad [1]$$

$$M = 2.01\times10^{30} \text{ kg} \quad [1]$$

mass of Sun = ... kg [2]

(iv) Explain why, although planets experience gravitational forces from the Sun, they do not fall into the Sun.

At any instant, each planet is moving in a direction tangential to its orbit.[1] According to Newton's first law, the planet wants to continue to move in this direction [1] instead of falling into the Sun.

[2]

[Total: 20]

9 **(a)** Distinguish between *electromotive force e.m.f.* and *potential difference p.d.* using energy considerations.

E.m.f. is the electrical energy converted from other forms of energy per unit charge delivered by the source. *P.d.* is the electrical energy converted to other forms of energy per unit charge when charges pass from one point to another. [1]

(b) Fig. 9.1 shows a variable resistor and a fixed resistor R_1 connected to a battery with e.m.f E and internal resistance r. The variable resistor takes a range of values between 0 Ω to R_{max}.

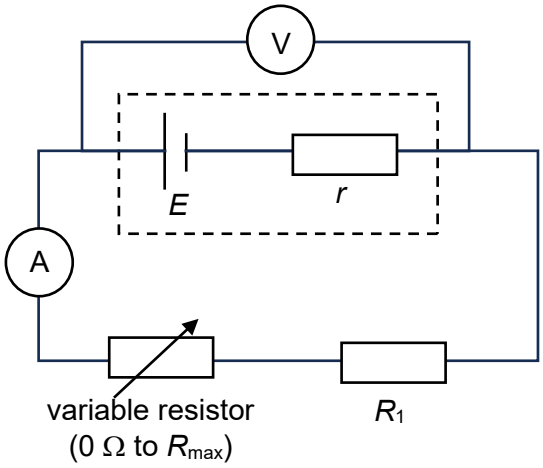


Fig. 9.1

The variation of the voltmeter reading V with current I through the ammeter as the variable resistor is varied over its **full range** of values is shown in Fig. 9.2.

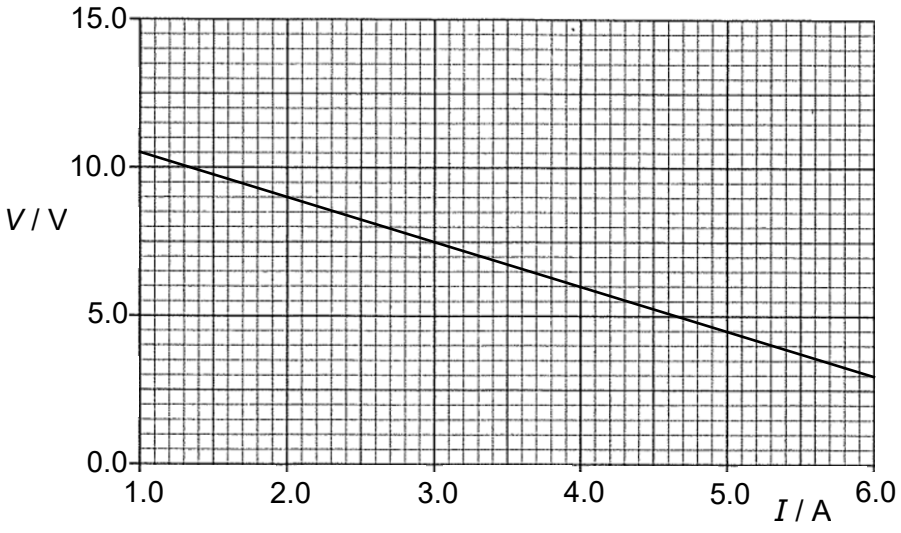


Fig. 9.2

(i) Using the graph, show that that internal resistance r is 1.5 Ω.

Since $E = V + Ir$ → $V = -rI + E$ (in the form of y = mx + c)

gradient of V–I graph gives **negative** value of internal resistance r

[1 identify which part of graph is related to r]

gradient = (10.5 – 3.0)/(1.0–6.0) [1 for correct coordinates on line, far apart]

gradient = –1.5 Ω

$\therefore r$ = 1.5 Ω [1]

[3]

(ii) Calculate the e.m.f. E of the battery.

When V = 10.5 V, I = 1.0 A

Using $E = V + Ir$

= 10.5 + 1.0(1.5) = 12.0 V

Note: E is **not** 10.5 V, as current at that point is not 0 A. Hence not y-intercept

e.m.f. E = 12.0 V [1]

(iii) Using Fig. 9.2, calculate the maximum resistance of the variable resistor R_{max}. Explain your working.

When variable resistor is set to 0 Ω, the resistance in the circuit is the lowest, hence current will be the highest. V = 3.0 V, I = 6.0 A

p.d. across R_1 is equal to V.

Using $V = IR_1$,

$R_1 = V / I$ = 3.0/6.0 = 0.50 Ω [1]

When variable resistor is set to max, the current will be the lowest.
V = 10.5 V, I = 1.0 A ,

Using $V = IR$, where R is the total resistance of R_{max} and R_1

$R_{max} + R_1 = V / I$

R_{max} + 0.50 = 10.5 / 1.0

R_{max} = 10.0 Ω [1]

(1 mark for explanation of choices of points)

resistance R_{max} = ... Ω [3]

(c) The variable resistor is now set to 6.0 Ω, and a thermistor is connected in parallel with the variable resistor and R_1. The ammeter is also now connected in series with the thermistor as shown in Fig. 9.3.

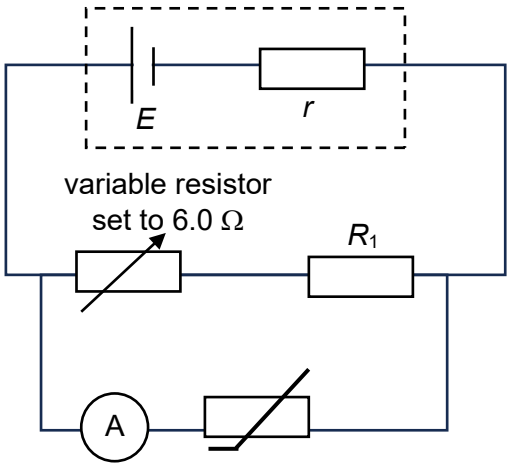


Fig. 9.3

(i) Explain how the potential difference across the thermistor changes when the temperature rises.

When the temperature increases, the resistance of the thermistor decreases. [1]

Hence the effective resistance of the external resistors and the thermistor, which are in parallel, decreases. By potential divider principle, the potential difference across both the external resistor and thermistor decreases. [1]

[2]

(ii) Hence, or otherwise, determine how the current across the thermistor changes when the temperature rises.

Since the effective resistance of the two external resistors and the thermistor decreases, the total resistance of the circuit decreases. Which means that the total current I_{total} exiting the cell increases.[1] Since the potential difference across the two external resistors decreases, by $V=IR$ the current I_R across the two external resistors decreases. [1]

Since $I_{total} = I_R + I_{Thermistor}$, the current through the thermistor increases. [1]

[3]

(d) Fig. 9.4 shows an alternating input voltage V_{IN} connected to a capacitor C and a resistor R.

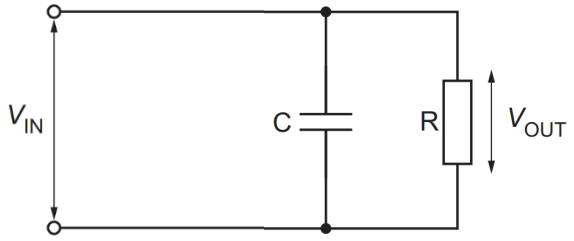


Fig. 9.4

The input voltage V_{IN} is a square wave. Fig 9.5 shows the variation of V_{IN} with time t.

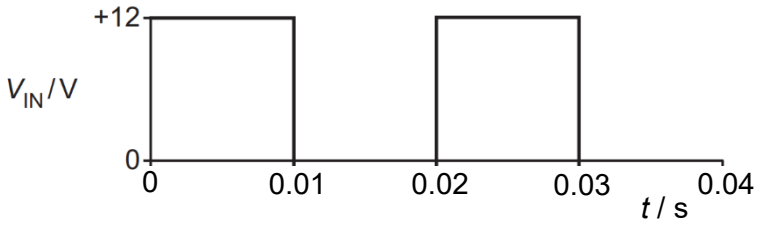


Fig. 9.5

Fig. 9.6 shows the variation of V_{OUT} with time t.

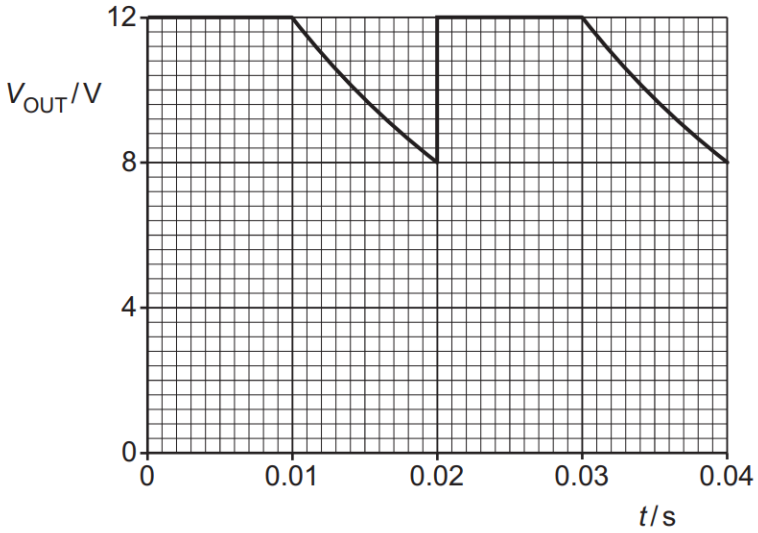


Fig. 9.6

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is 570 μF.

Energy stored in a capacitor $E = \frac{1}{2}CV^2$

$C = 2E/V^2 = \frac{2 \times 0.041}{12^2}$ [1]

$C = 569.44 \times 10^{-6}$ [1]

$= 570\ \mu F$

[2]

(ii) Determine the resistance of R.

It takes 0.010 s for p.d. to drop from 12.0 V to 8.0 V

Using $V = V_0 e^{-\frac{t}{\tau}}$, where $\tau = RC$

$8.0 = (12.0)\, e^{-\frac{0.010}{RC}}$ [1]

$\ln(8.0/12.0) = -\dfrac{0.010}{R \times 570 \times 10^{-6}}$

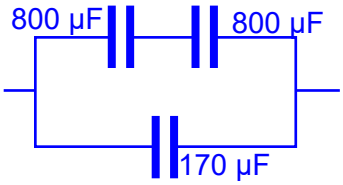
$R = 43.27\ \Omega = 43\ \Omega$ [1]

(note1: while not a marking pt, time is read to half a square precision, 0.010s)

(note2: recommended to use the proven value of C)

resistance = ... Ω [2]

(iii) Given two capacitors rated 800 μF and a third capacitor of 170 μF, suggest how capacitor *C* of 570 μF can be constructed from these three capacitors. Include a labelled diagram and show your workings clearly.



1m for diagram

Capacitance of two 800 μF in series = $\left(\dfrac{1}{800} + \dfrac{1}{800}\right)^{-1} = 400$ μF [1]

Capacitance of 400 μF and 170 μF in parallel = 400 + 170 = 570 μF [1]

[3]

[Total: 20]